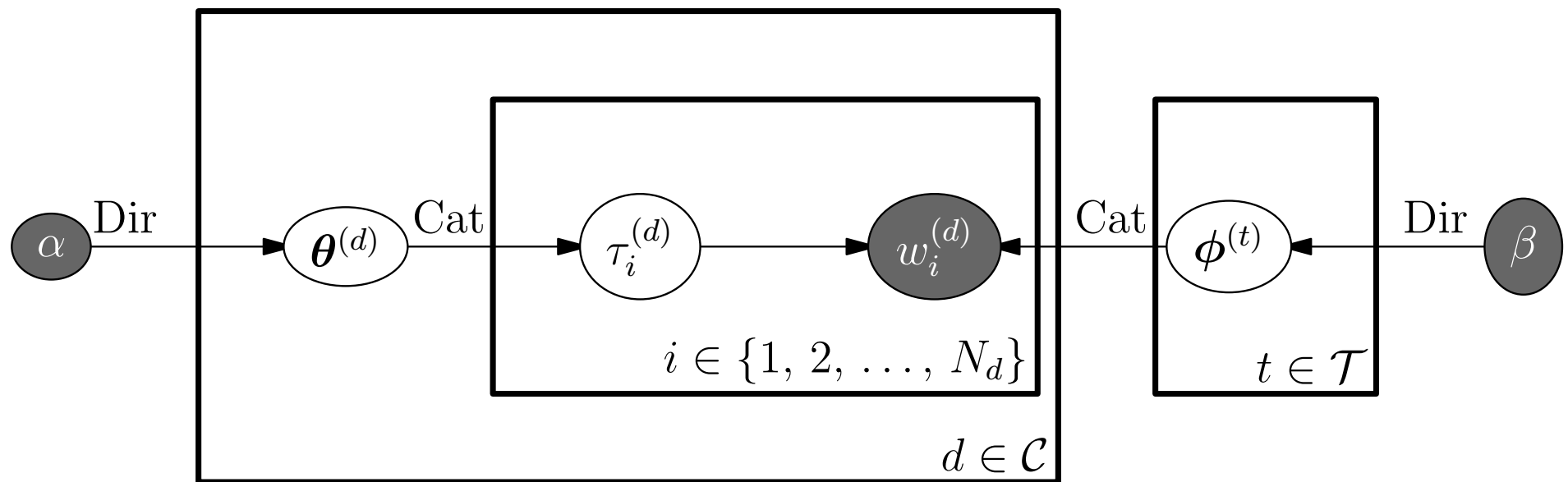


Advanced Machine Learning

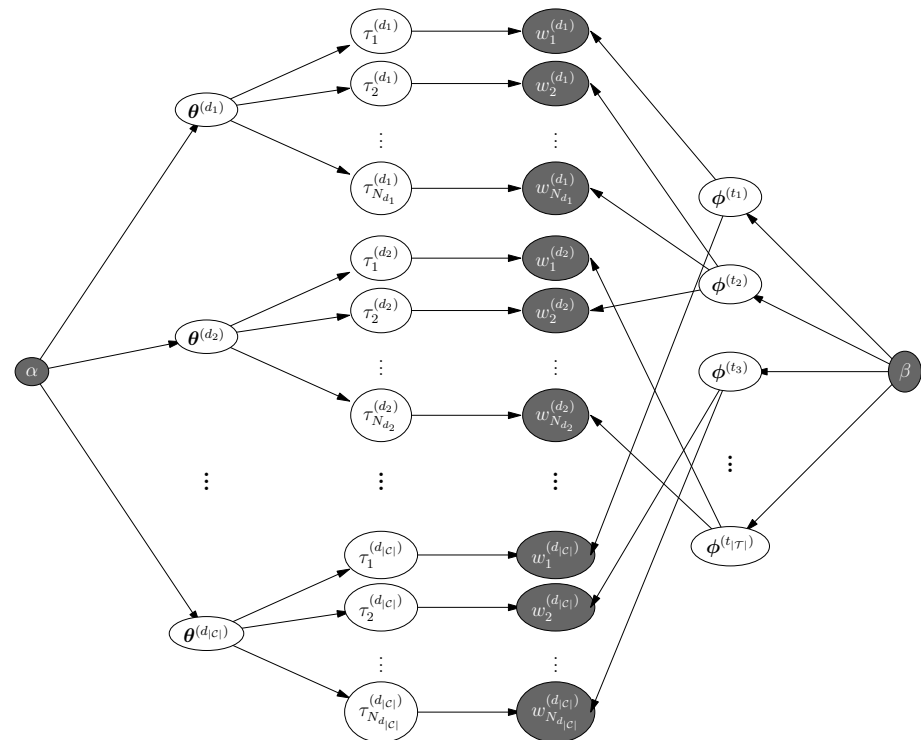
Graphical Models



Conditional Independence, Graphical models, LDA

Outline

1. Graphical Models
2. Cakes!
3. Latent Dirichlet Allocation



Graphical Models

- If we want to build large probabilistic inference systems

- ★ AI Doctor

- ★ Fault diagnostic system for a computer

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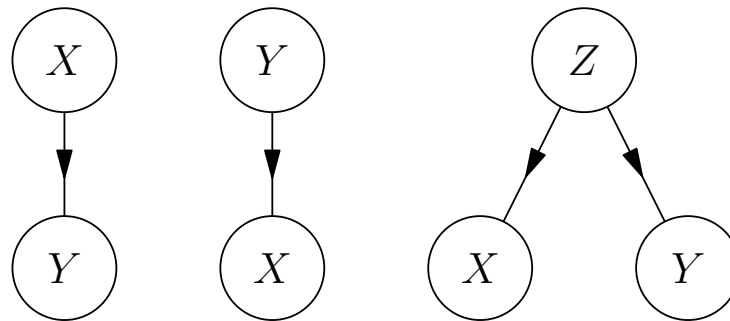
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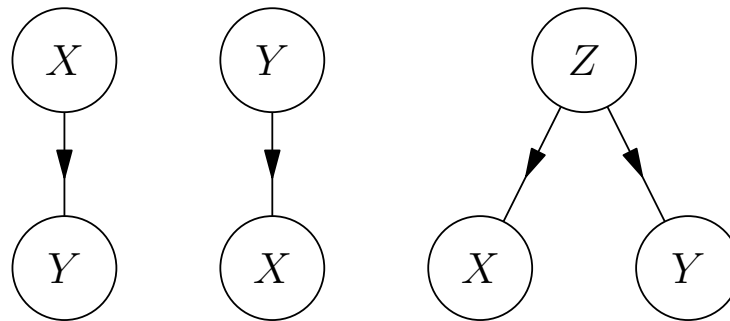
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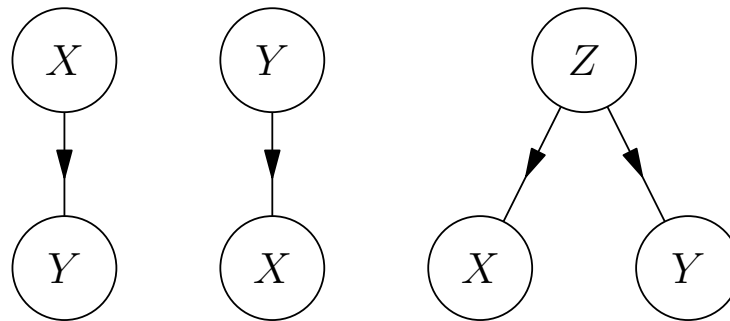
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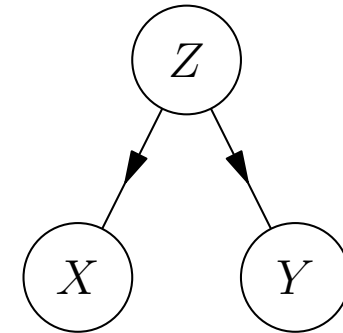
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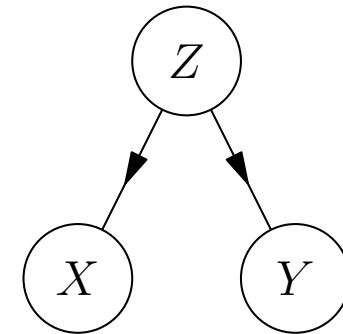
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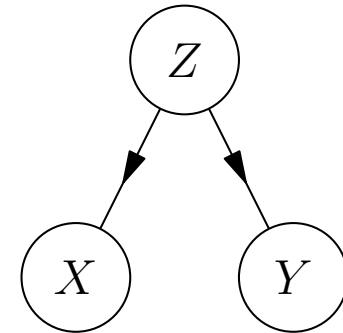
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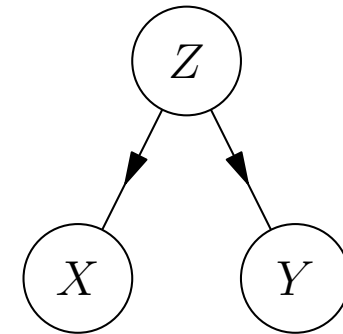
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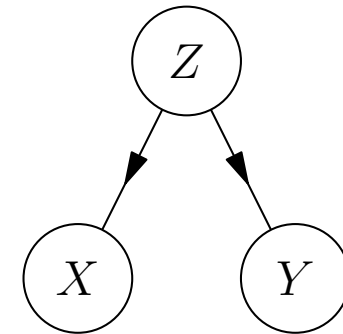
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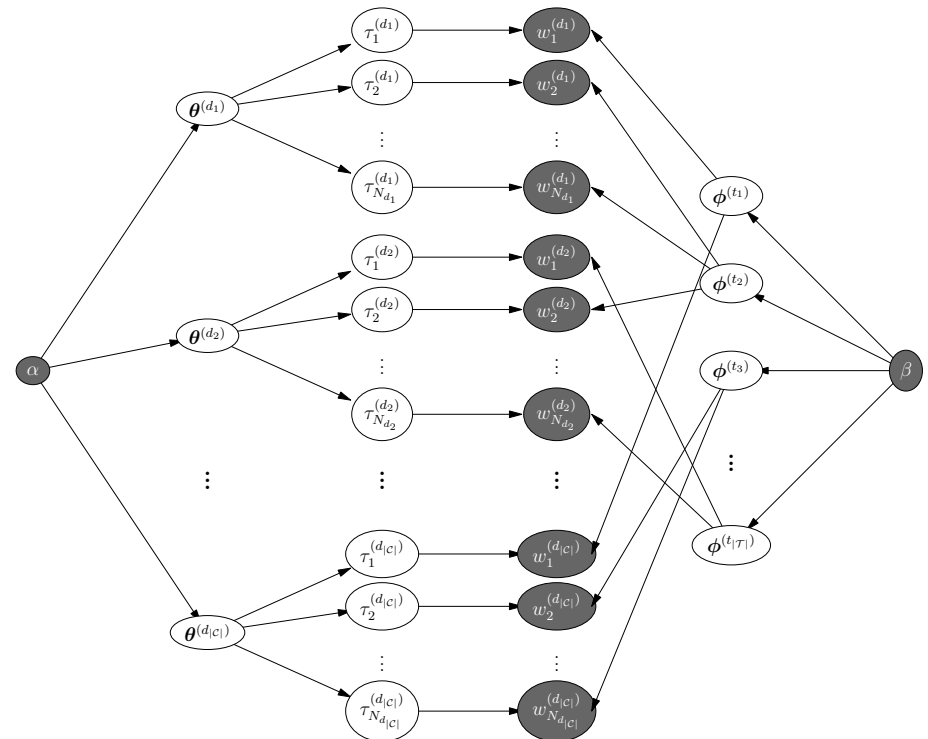


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- I will go through a very simple example involving cakes
- It illustrates some simple principles
- In the subsidiary notes I present a very simple program for computing all the probabilities

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The Cake Scenario

- Abi and Ben both bake cakes and bring them into the coffee room
- Abi will bring in cakes 20% of the time: $\mathbb{P}(A = 1) = 0.2$
- Ben will bring in cakes 10% of the time: $\mathbb{P}(B = 1) = 0.1$
- 90% of the time if either Abi or Ben have put cakes in the coffee room there is some left when I enter
$$\mathbb{P}(C = 1|A = 1, B = 0) = \mathbb{P}(C = 1|A = 0, B = 1) = 0.9$$
- If they both make cake then there is always cake left
$$\mathbb{P}(C = 1|A = 1, B = 1) = 1$$
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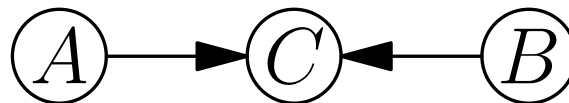
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- Other probabilities I can deduce, e.g.

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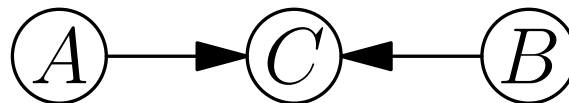
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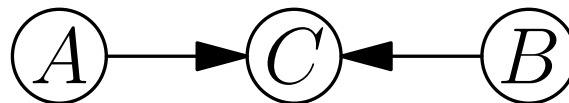
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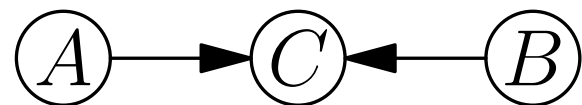
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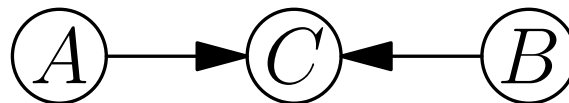
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Computing Expectations

- By using the joint probability and summing over all unknown quantities, we can compute expectations of anything we are interested in
- These sums are often sped up using knowledge of conditional independence
- To compute the probability of an event $\mathcal{E}$ we introduce an indicator function $\llbracket \mathcal{E} \rrbracket$ which is equal to 1 if the event happens and 0 otherwise

$$\mathbb{P}(\mathcal{E}) = \mathbb{E}[\llbracket \mathcal{E} \rrbracket]$$

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Are There Any Cakes Left?

- We can use our model to compute the probabilities of there being cakes in the coffee room

$$\begin{aligned}\mathbb{P}(C = 1) &= \sum_{A,B,C \in \{0,1\}} \mathbb{I}[C = 1] \mathbb{P}(A,B,C) \\ &= \sum_{A,B \in \{0,1\}} \mathbb{P}(C = 1|A,B) \mathbb{P}(A) \mathbb{P}(B) = 0.29\end{aligned}$$

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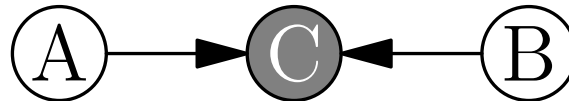
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- In graphical models observed random variables are shaded



- The probabilities conditioned on C is given by

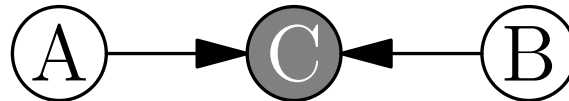
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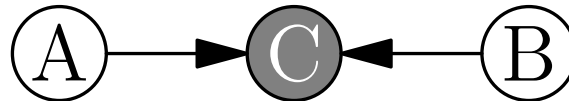
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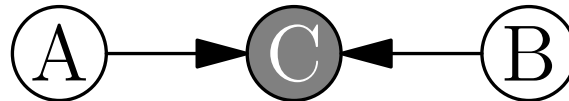
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Who Made Those Cakes?

- If we observe there are cakes

$$\mathbb{P}(A, B | C = 1) = \mathbb{P}(A, B, C = 1) / \mathbb{P}(C = 1)$$

- A straightforward if tedious calculation shows

$$\mathbb{P}(A = 1 | C = 1) = 0.628, \quad \mathbb{P}(B = 1 | C = 1) = 0.317$$

$$\mathbb{P}(A = 1, B = 1 | C = 1) = 0.069$$

- Note $\mathbb{P}(A = 1, B = 1 | C = 1) \neq \mathbb{P}(A = 1 | C = 1) \mathbb{P}(B = 1 | C = 1)$
- When we observe C then A and B are no longer independent

Who Made Those Cakes?

- If we observe there are cakes

$$\mathbb{P}(A, B | C = 1) = \mathbb{P}(A, B, C = 1) / \mathbb{P}(C = 1)$$

- A straightforward if tedious calculation shows

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Elaborate Cakes

- We can elaborate on our cake model
- We suppose that Dave likes cakes so if there is a cake in the coffee room there is a 80% chance that I will see him eating a cake: $\mathbb{P}(D = 1|C = 1) = 0.8$
- Even if there are no cakes in the coffee room there is a 10% chance that Dave has bought his own cake:
 $\mathbb{P}(D = 1|C = 0) = 0.1$
- Eli also likes cakes: there is a 60% chance that I will see her eating cakes if there are cakes in the coffee room:
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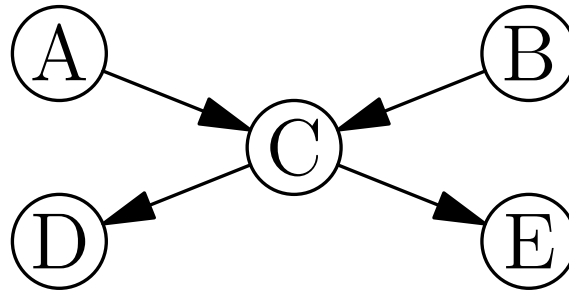
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- We can depict this situation as



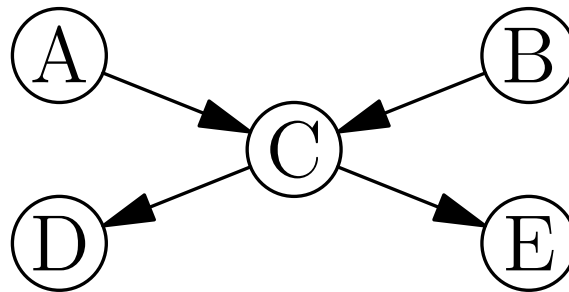
- This allows us to break down the joint probability

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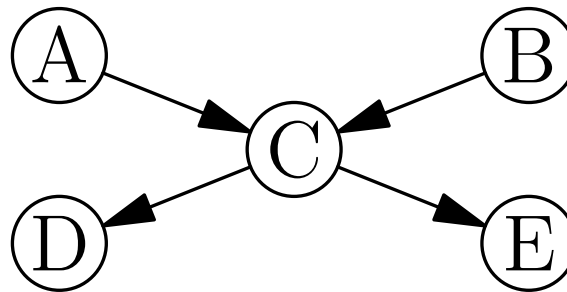
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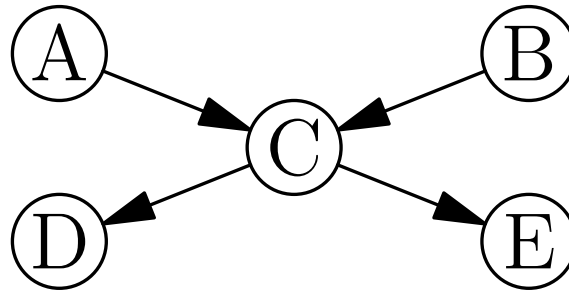
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Dependencies

- If we don't observe cakes then the probability of Dave and Eli eating cake are not independent

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$$\mathbb{P}(D = 1, E = 1) = 0.1392$$

- This changes if we know there are cakes in the coffee room

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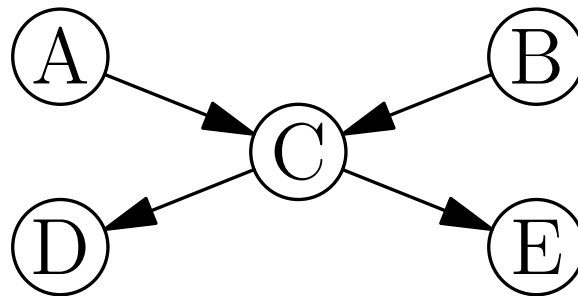
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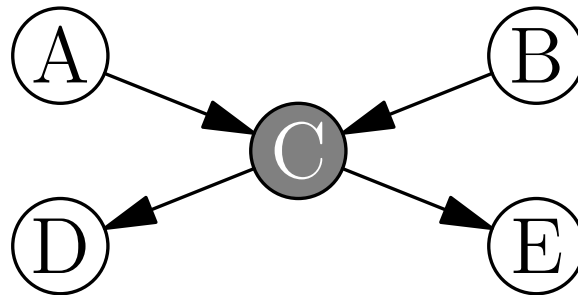


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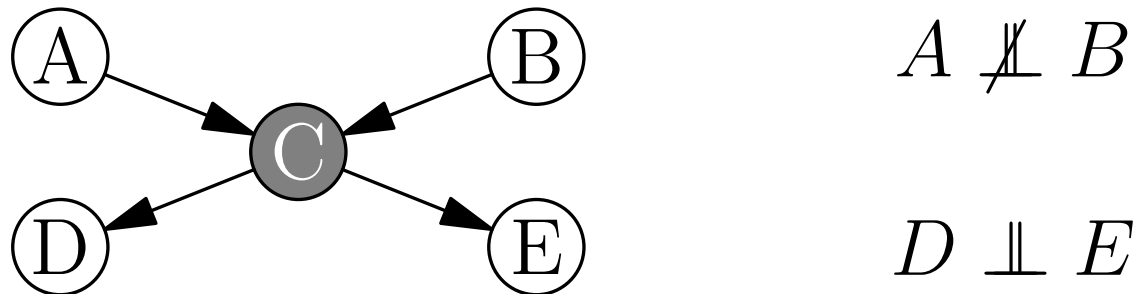


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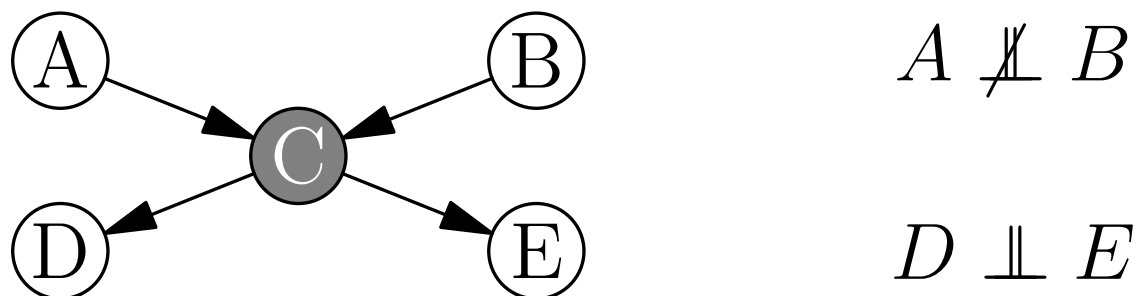
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- There are rules to deduce the conditional independence from a graphical model given which variables have been observed—but these are details that you can look up if needed

Graphical Model Frameworks

- There are sophisticated frameworks for computing probabilities in Bayesian Belief Networks efficiently
- If our graph is a tree then we can evaluate probabilities efficiently
- When there are loops (so that a random variable both influences and is influenced by another random variables) then exact evaluation of expectations requires exhaustive summing over variables
- There are various message passing algorithms designed to obtain approximations of expectations

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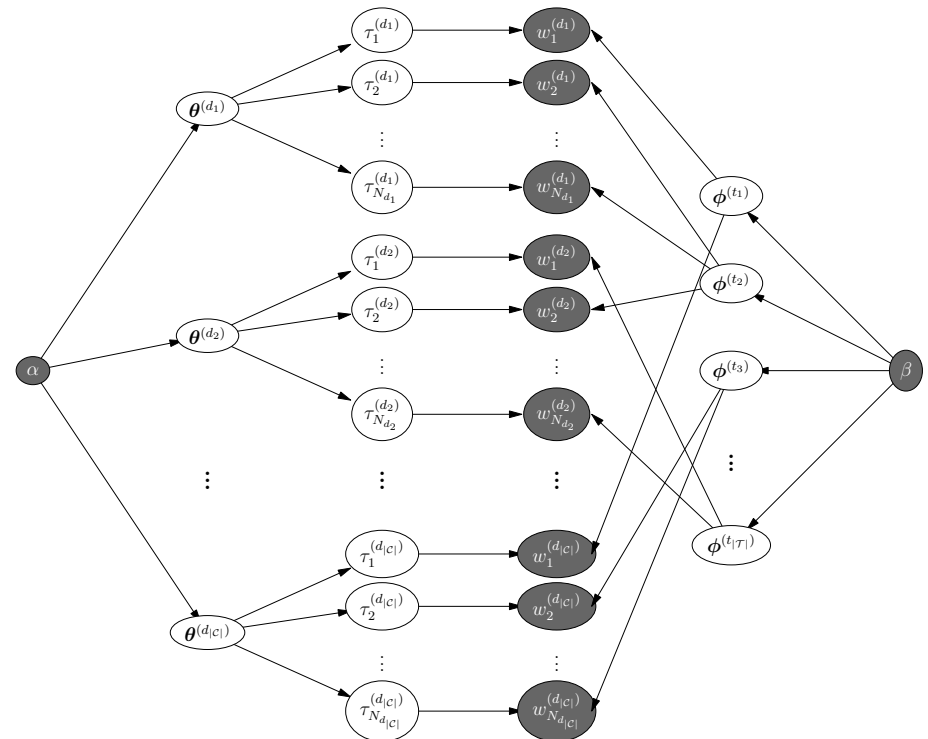
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Outline

1. Graphical Models
2. Cakes!
3. **Latent Dirichlet Allocation**



Model for Documents

- We consider a model for the words in a set of documents (we ignore word order)
- We consider a corpus $\mathcal{C} = \{d_i | i = 1, 2, \dots, |\mathcal{C}|\}$
- With documents consisting of words

$$d = \left(w_1^{(d)}, w_2^{(d)}, \dots, w_{N_d}^{(d)} \right)$$

- We assume that there is a set of topics $\mathcal{T} = \{t_1, t_2, \dots, t_{|\mathcal{T}|}\}$
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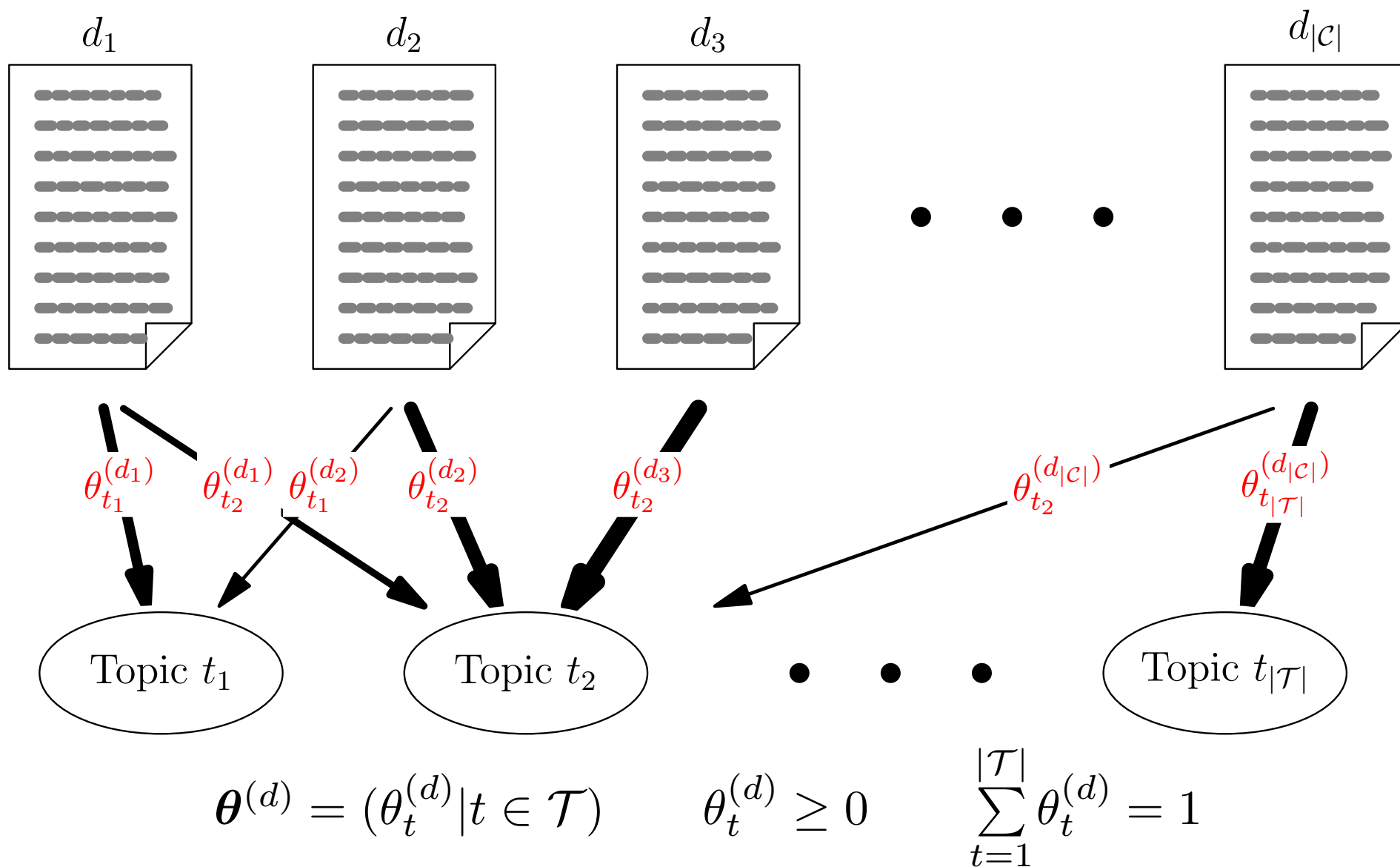
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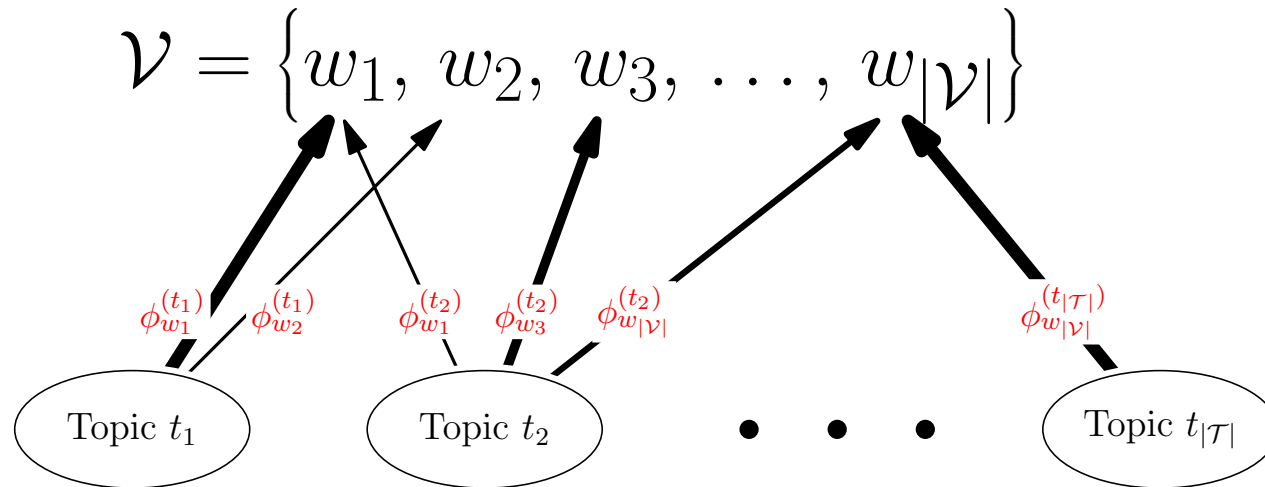
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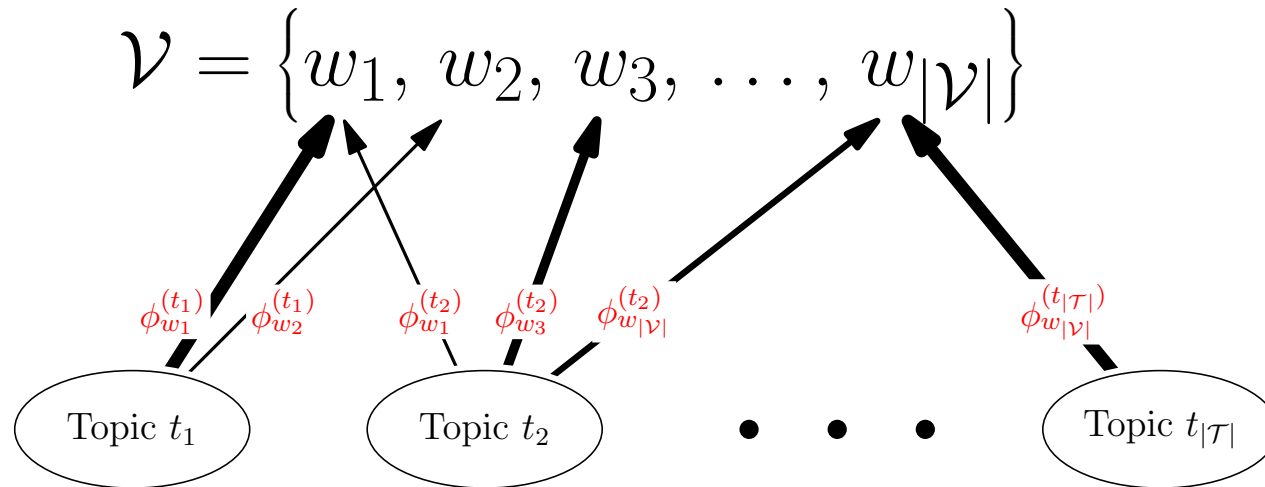
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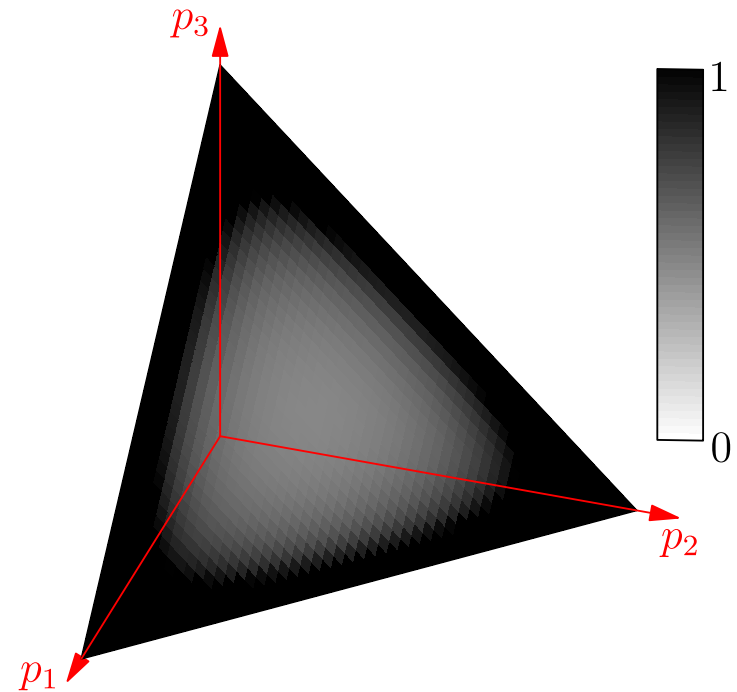
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Dirichlet Allocation

- Most documents are predominantly about a few topics and most topics have a small number of words associated to them
- We can generate sparse vectors $\theta^{(d)}$ and $\phi^{(t)}$ from a Dirichlet distribution with small parameters α

$$\text{Dir}(\mathbf{p}|\boldsymbol{\alpha}) = \Gamma\left(\sum_i \alpha_i\right) \prod_{i=1}^n \frac{p_i^{\alpha_i-1}}{\Gamma(\alpha_i)}$$

- $\sum_i p_i = 1$



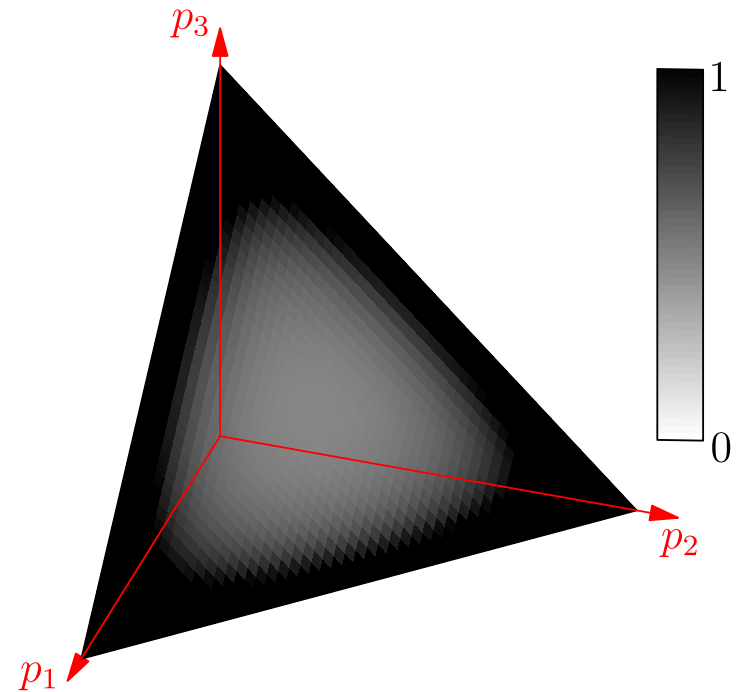
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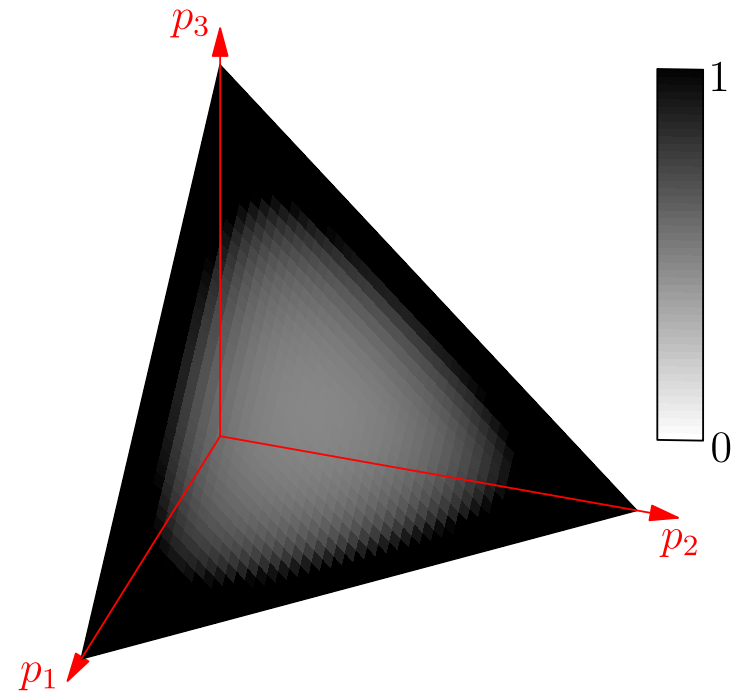
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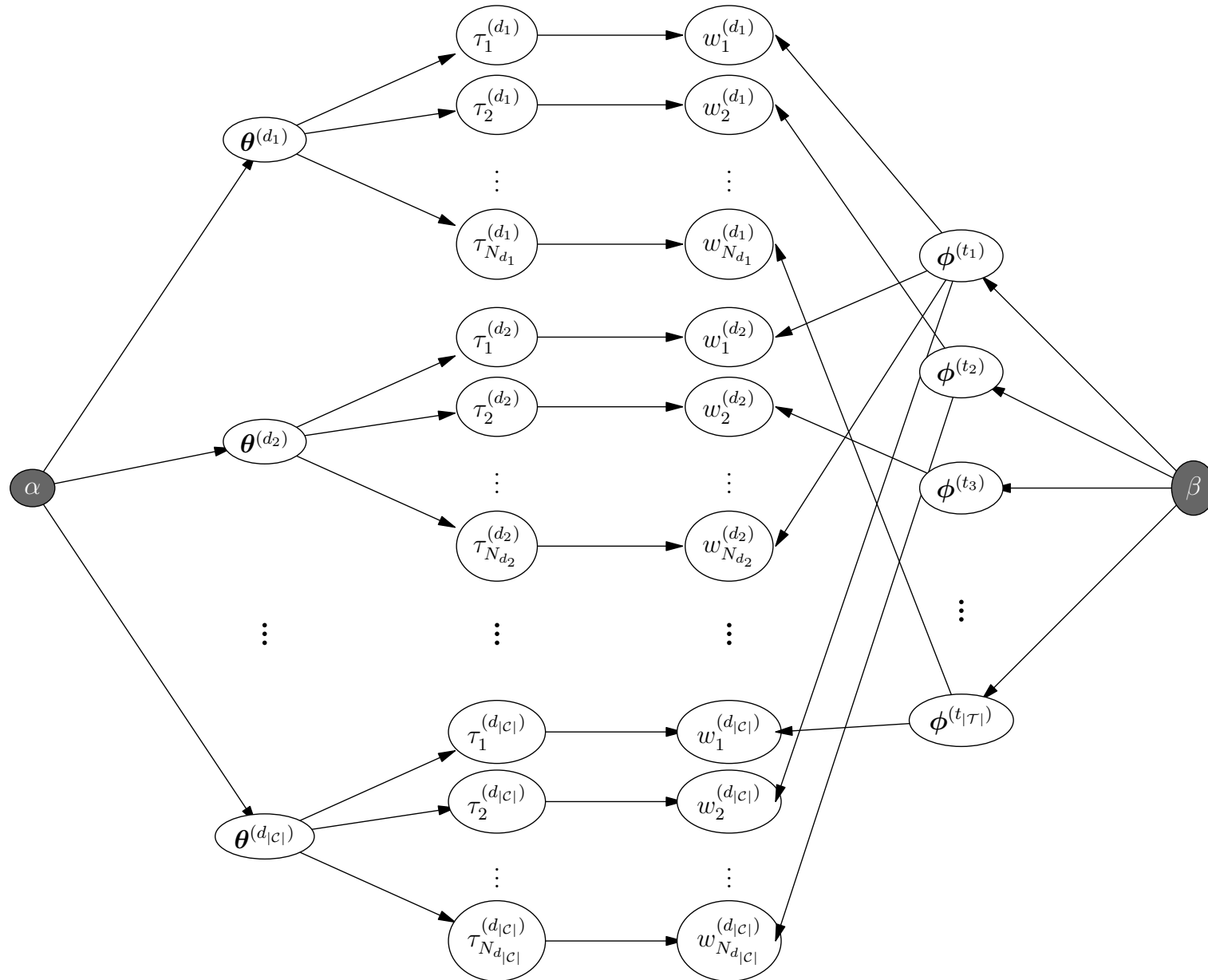
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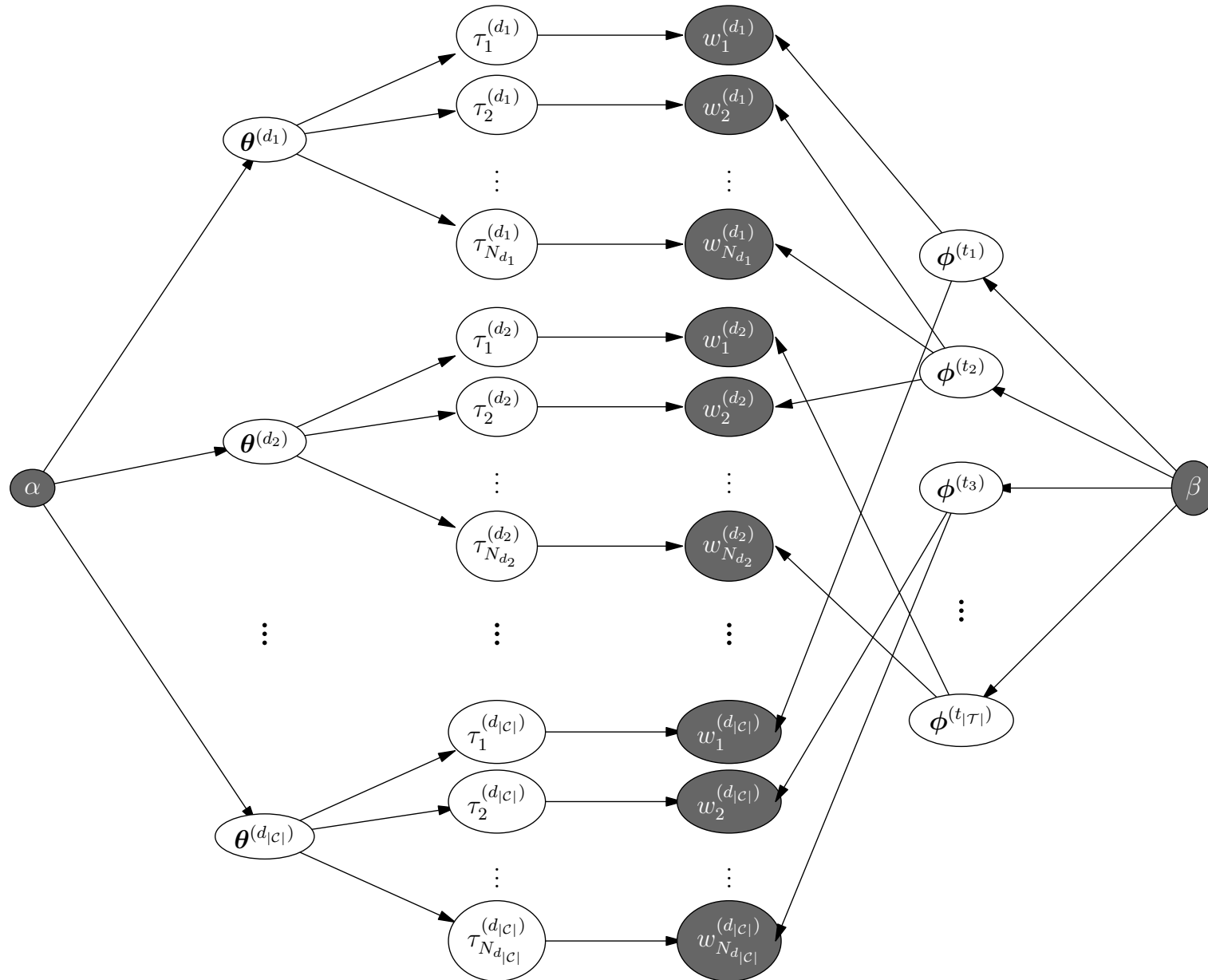
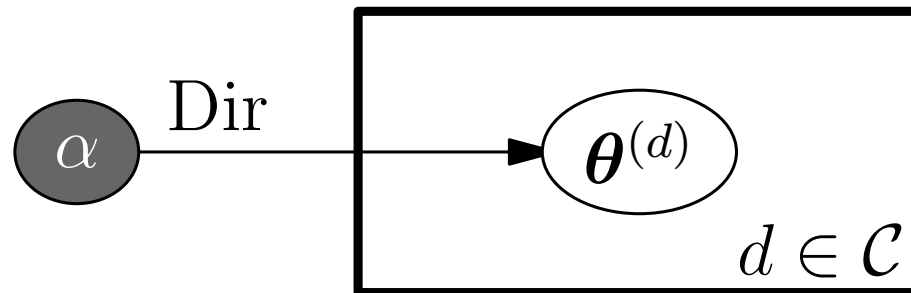


Plate Diagrams

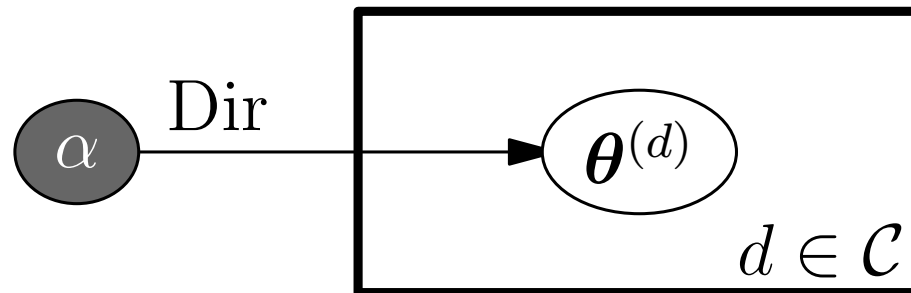
- Drawing every random variable is tedious (and not really possible)
- A short-hand is to draw a box (plate) meaning repeat



- That is we generate vectors θ^d from a Dirchelet distribution $\text{Dir}(\theta|\alpha\mathbf{1})$ for all documents in corpus $\mathcal{C}$

Plate Diagrams

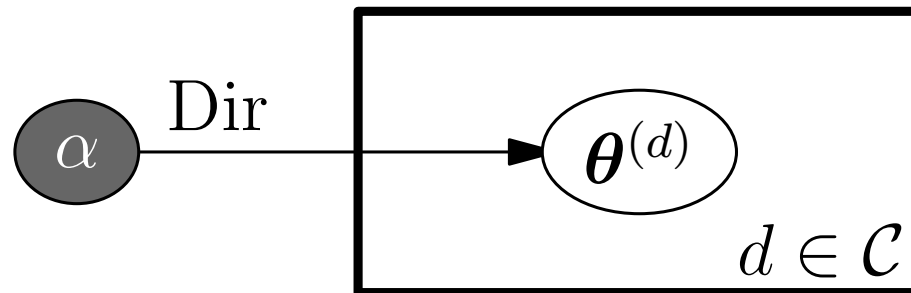
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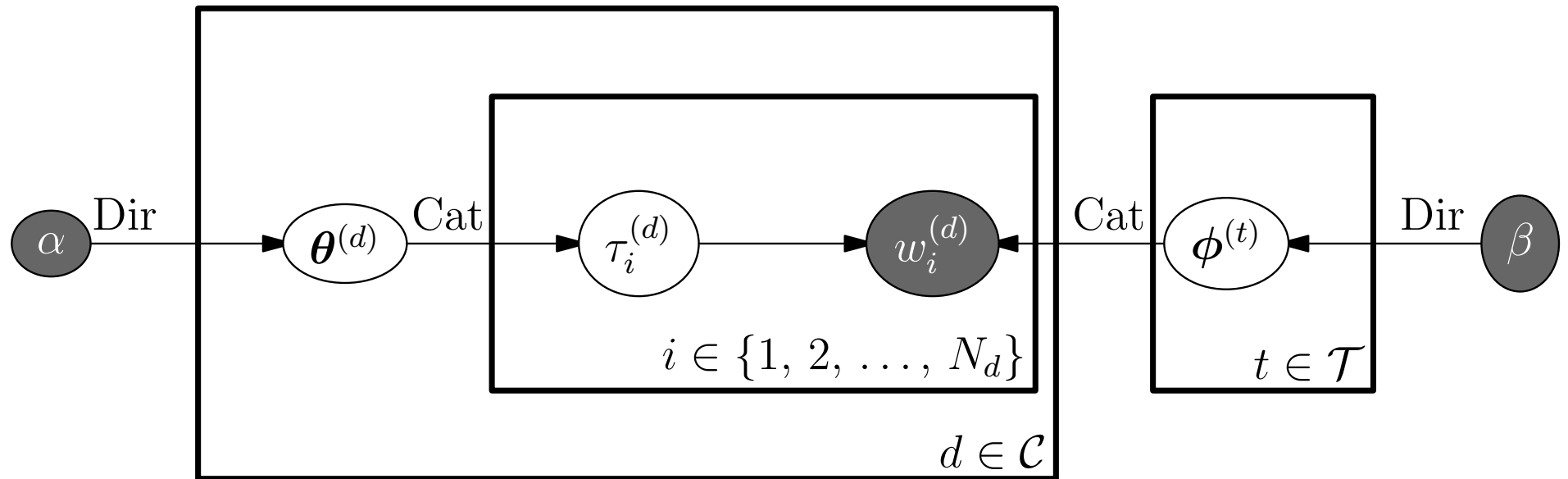
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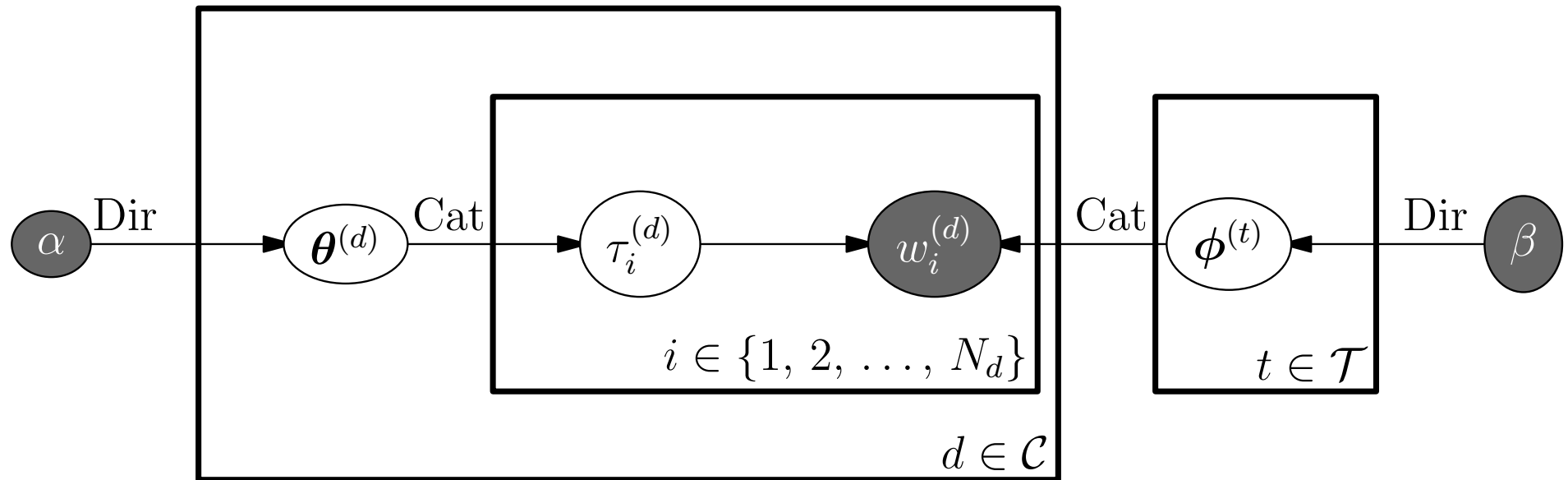
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Probabilistic Model

- The graphical Model is shorthand for the variables

$$\mathbf{W} = (\mathbf{w}^{(d)} | d \in \mathcal{C}) \quad \text{with} \quad \mathbf{w}^{(d)} = (w_1^{(d)}, w_2^{(d)}, \dots, w_{N_d}^{(d)}), \quad \text{and} \quad w_i^{(d)} \in \mathcal{V}$$

$$\mathbf{T} = (\tau_i^{(d)} | d \in \mathcal{C} \wedge i \in \{1, 2, \dots, N_d\}) \quad \text{with} \quad \tau_i^{(d)} \in \mathcal{T}$$

$$\mathbf{\Theta} = (\boldsymbol{\theta}^{(d)} | d \in \mathcal{C}) \quad \text{with} \quad \boldsymbol{\theta}^{(d)} = (\theta_t^{(d)} | t \in \mathcal{T}) \in \Lambda^{|\mathcal{T}|}$$

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- Distributed according to

$$\mathbb{P}(\mathbf{W}, \mathbf{T}, \mathbf{\Theta}, \mathbf{\Phi} | \alpha, \beta) = \left(\prod_{t \in \mathcal{T}} \text{Dir}(\phi^{(t)} | \beta \mathbf{1}) \right) \left(\prod_{d \in \mathcal{C}} \text{Dir}(\boldsymbol{\theta}^{(d)} | \alpha \mathbf{1}) \prod_{i=1}^{N_d} \text{Cat}(\tau_i^{(d)} | \boldsymbol{\theta}^{(d)}) \text{Cat}(w_i^{(d)} | \phi^{(\tau_i^{(d)})}) \right)$$

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Finding Topics

- We are given the set of words W and don't really care about τ_i^d the topic associated with word i in document d
- But we are interested in the words associated with each topic $\phi^{(t_i)}$
- And the topics associated with each document $\theta^{(d)}$
- To compute them we need to sample the probability distribution
- One way to do this is using Monte Carlo methods (see next lecture)

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Summary

- Building probabilistic models is an intricate process
- Graphical models provide a representation showing the causal relationship between random variables
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